Explainable dyslexia analysis on EDTD70: PPEI, LEPI, RFI, OPEI

Nested leave-one-participant-out evaluation; selected model: **LogisticRegression**; exact Shapley attribution.

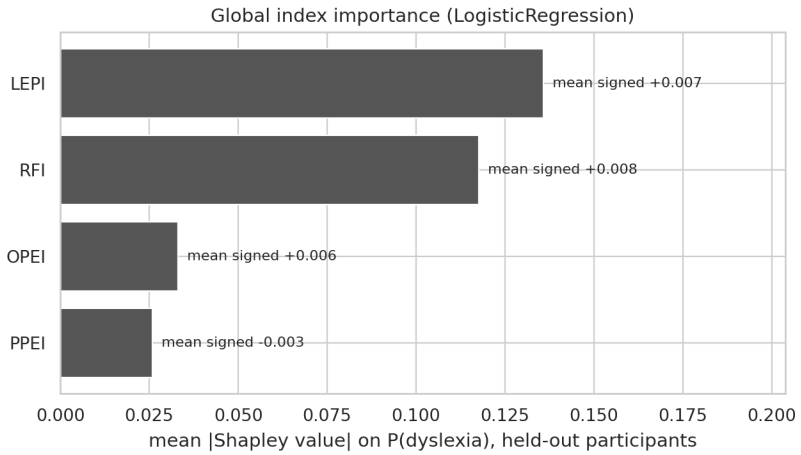
Held-out performance (95% bootstrap CI)

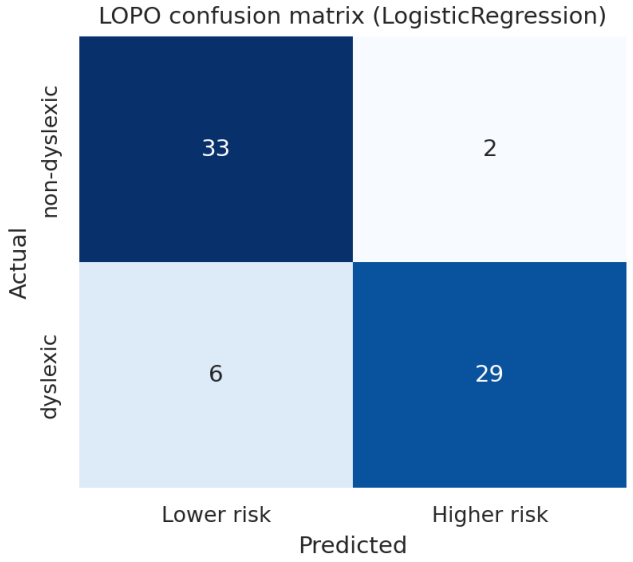
Metric	Value	95% CI
accuracy	0.886	[0.814, 0.957]
precision	0.935	[0.840, 1.000]
recall_sensitivity	0.829	[0.692, 0.944]
specificity	0.943	[0.861, 1.000]
f1	0.879	[0.784, 0.952]
roc_auc	0.915	[0.837, 0.979]

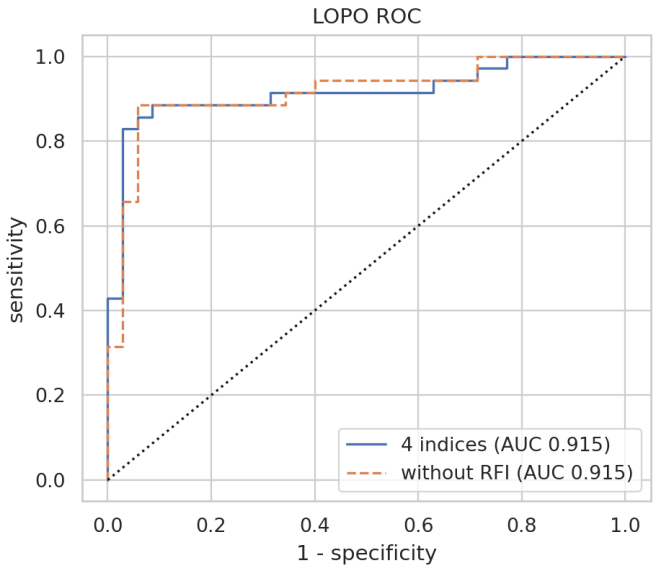
Confusion matrix: TN=33 FP=2 FN=6 TP=29. Sensitivity analysis without RFI (its weights were fitted on all 70 labels): accuracy 0.900, ROC-AUC 0.915.

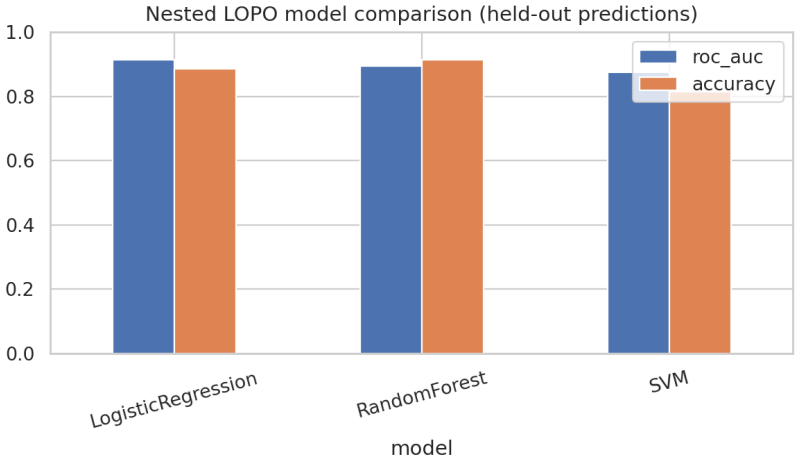
Global index ranking (mean |Shapley|)

Rank	Index	Mean |SHAP|	Mean signed	Main index (n)
1	LEPI	0.1359	+0.0073	15
2	RFI	0.1177	+0.0082	15
3	OPEI	0.0331	+0.0064	8
4	PPEI	0.0258	-0.0026	11




LOPO confusion matrix (LogisticRegression)
Actual
non-dyslexic
dyslexic
33
2
6
29
Lower risk
Higher risk
Predicted


LOPO ROC
sensitivity
1 - specificity
0.0
0.2
0.4
0.6
0.8
1.0
4 indices (AUC 0.915)
without RFI (AUC 0.915)


Nested LOPO model comparison (held-out predictions)
roc_auc
accuracy
LogisticRegression
RandomForest
SVM
model

Participant table (primary output)

Participant	PPEI	LEPI	RFI	OPEI	Predicted result	Main contributing index
1003	1.91583	0.13574	-3.9733	0.01326	Lower risk	None identified
1009	0.69564	0.22954	-1.9434	0.00881	Lower risk	OPEI
1016	2.03966	0.19540	-1.9293	0.00858	Lower risk	OPEI
1019	-0.4661	0.17972	-3.4281	0.01173	Lower risk	None identified
1021	0.10925	0.21208	-2.2089	0.00918	Higher risk	OPEI
1033	3.70013	0.12744	-6.2142	0.01905	Lower risk	None identified
1038	-3.0433	0.48706	4.96363	0.00276	Higher risk	RFI
1040	0.41831	0.18110	-3.6295	0.01213	Lower risk	None identified
1058	-0.0546	0.13313	-3.9436	0.01245	Lower risk	None identified
1065	0.57984	0.23299	-1.9111	0.00876	Higher risk	OPEI
1073	0.32205	0.09782	-4.2145	0.01375	Lower risk	None identified
1075	2.27344	0.14260	-3.6983	0.01425	Lower risk	None identified
1082	-1.7722	0.35513	1.31258	0.00645	Higher risk	RFI
1090	2.89101	0.15390	-4.4311	0.01194	Lower risk	None identified
1095	1.44619	0.11443	-4.2226	0.01744	Lower risk	None identified
1109	3.14448	0.05124	-6.1808	0.01940	Lower risk	None identified
1113	-1.8509	0.32424	0.66678	0.00601	Higher risk	RFI
1115	2.48194	0.13697	-4.5249	0.01473	Lower risk	None identified
1134	-3.1905	0.37964	0.62126	0.00598	Higher risk	LEPI
1145	-0.3726	0.12300	-3.4903	0.01253	Lower risk	PPEI
1160	1.98731	0.11279	-4.2928	0.01653	Lower risk	None identified
1166	-2.1623	0.39056	2.11098	0.00475	Higher risk	RFI
1169	3.20327	0.08218	-5.0597	0.01652	Lower risk	None identified
1174	0.77613	0.21829	-2.7611	0.01019	Lower risk	LEPI
1186	2.66286	0.07809	-4.9074	0.01493	Lower risk	PPEI
1187	4.65229	0.07431	-7.6190	0.02018	Lower risk	PPEI
1189	1.61844	0.16175	-3.1148	0.01246	Lower risk	PPEI
1209	1.57995	0.16292	-2.2218	0.00860	Lower risk	OPEI
1235	4.43866	0.08483	-7.4179	0.02068	Lower risk	PPEI
1254	-1.7379	0.17030	-1.7576	0.00880	Lower risk	RFI
1255	1.32559	0.12615	-3.3726	0.01707	Lower risk	OPEI
1257	0.45335	0.10550	-4.3844	0.01618	Lower risk	None identified
1258	1.79387	0.17703	-3.7028	0.01685	Lower risk	OPEI
1263	1.49596	0.09548	-4.7233	0.01465	Lower risk	PPEI
1271	3.65970	0.09823	-5.3541	0.01456	Lower risk	PPEI

1274	3.67355	0.11983	-5.9882	0.01644	Lower risk	PPEI
1284	1.64022	0.15769	-3.7290	0.01389	Lower risk	PPEI
1300	-2.8633	0.48025	2.43185	0.00397	Higher risk	LEPI
1312	-1.3833	0.37364	1.71638	0.00404	Higher risk	RFI
1314	0.66351	0.13257	-3.2138	0.01164	Lower risk	None identified
1318	-2.0163	0.29474	-0.0438	0.00714	Higher risk	LEPI
1322	1.91300	0.18908	-2.6238	0.00982	Lower risk	OPEI
1345	5.99508	0.02510	-9.3784	0.02854	Lower risk	PPEI
1349	-0.0554	0.22481	-1.2250	0.00892	Higher risk	RFI
1350	-4.0566	0.40979	2.12999	0.00386	Higher risk	LEPI
1377	3.99850	0.06796	-6.4010	0.01607	Lower risk	None identified
1380	3.92152	0.15799	-6.5907	0.01411	Lower risk	None identified
1398	4.33442	0.08704	-6.5981	0.01769	Lower risk	None identified
1405	1.50942	0.20902	-2.1785	0.00847	Higher risk	RFI
1417	2.13664	0.16015	-3.7449	0.01591	Lower risk	None identified
1421	-2.9092	0.38278	0.68148	0.00502	Higher risk	LEPI
1459	-6.2312	0.61540	4.10439	0.00352	Higher risk	LEPI
1476	-1.3628	0.34301	1.53015	0.00469	Higher risk	LEPI
1571	-1.5200	0.30200	0.85694	0.00560	Higher risk	RFI
1582	-2.3123	0.36326	0.33426	0.00633	Higher risk	LEPI
1591	-2.2901	0.34640	0.93747	0.00664	Higher risk	LEPI
1626	-3.0252	0.21976	-0.8471	0.00746	Higher risk	RFI
1693	-4.9923	0.33238	-0.1235	0.00549	Higher risk	PPEI
1729	-4.9093	0.79661	6.32837	0.00242	Higher risk	LEPI
1744	0.88394	0.20763	-1.9367	0.00967	Higher risk	RFI
1760	0.29123	0.12371	-3.8372	0.01528	Lower risk	None identified
1858	-8.4153	0.92877	3.66027	0.00368	Higher risk	LEPI
1859	-6.8779	0.80288	3.76388	0.00238	Higher risk	LEPI
1869	2.86514	0.14381	-3.6462	0.01114	Lower risk	None identified
1879	-3.5021	0.44574	3.15471	0.00322	Higher risk	RFI
1903	-4.3948	0.60665	3.97715	0.00393	Higher risk	LEPI
1913	-3.0426	0.40873	1.49222	0.00462	Higher risk	RFI
1929	-5.4023	0.56768	3.52812	0.00308	Higher risk	LEPI
1993	0.79717	0.24242	-0.5138	0.00762	Higher risk	RFI
1996	-1.0443	0.29219	-0.7048	0.00646	Higher risk	RFI

Limitations: 70 participants; exploratory research indices, not a clinical diagnostic; Shapley values describe the model, not the cause of dyslexia; indices are correlated; RFI constants carry label information from all participants.